DO NOW — RECALL

Boolean algebra rules

Name: _____ Class: _____ Date: _____

1. Normalise 0.0110×2^4 using a two’s-complement mantissa. Give the new mantissa and exponent.

2. A sorted array is [3, 8, 12, 19, 27, 35, 42] . When searching for 27 , which indexes are checked, in order?

3. Show the array after one full left-to-right pass of ascending bubble sort on [7, 3, 5, 2] .

4. An alarm sounds if the door is open AND the system is armed, OR if panic is pressed. Write Q using A, B and C.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

0.1100×2^3 . Shift the mantissa one place left and decrease the exponent by one to preserve the value. A normalised positive two’s-complement mantissa begins 01 ; a negative one begins 10 .

2. Model answer

Indexes 3 → 5 → 4 (values 19 → 35 → 27). Binary search tests the middle index of the remaining sorted range, using zero-based indexing.

3. Model answer

[3, 5, 2, 7] . Compare adjacent values and swap when the left value is larger. The largest value, 7, moves to the end.

4. Model answer

$Q = (A \wedge B) \vee C$. A = door open, B = armed, C = panic pressed. Group door AND armed first; panic is an alternative trigger.

By the end, you can…

• apply De Morgan's Laws and double negation

• use distribution, association and commutation to rearrange expressions

• show each algebra step clearly in OCR-style simplification

Vocabulary — know these terms

Recall the term — De Morgan

Recall the term — distribution

Recall the term — association

Recall the term — commutation

Recall the term — double negation

Recall the term — simplify

Recall the term — equivalent

Recall the term — brackets

NEW LEARNING

Read, annotate and explain

Change form without changing truth

De Morgan:
¬(A ∧ B) = ¬A ∨ ¬B
¬(A ∨ B) = ¬A ∧ ¬B

Double negation:
¬¬A = A

Name the rule, apply one change at a time, and keep the expression logically equivalent.

De Morgan

The operator inside the brackets swaps as each term is negated.

Double negation

Two consecutive NOT operations cancel.

Distributive law

A appears in both groups. Write it once outside the brackets, then join B and C with OR.

The matching OR form is also useful: (A ∨ B) ∧ (A ∨ C) = A ∨ (B ∧ C).

Association + commutation

These allow regrouping and reordering before other simplification.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Simplify ¬(A ∧ B)

Outer NOT over AND

Use De Morgan's Law.

Negate each term

A → ¬A
B → ¬B

AND becomes OR

¬A ∨ ¬B

Equivalent behaviour

Both are false only when A=1 and B=1.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. De Morgan

Simplify ¬A ∨ ¬B using the relevant De Morgan form.

```
¬A ∨ ¬B
= ¬(? ? ?)
```

Method / working:

2. Double negation

Simplify ¬(¬B).

```
¬(¬B)
= ?
```

Method / working:

3. Distribution

Simplify (A ∨ B) ∧ (A ∨ C).

```
(A ∨ B) ∧ (A ∨ C)
= A ∨ (? ? ?)
```

Method / working:

4. Commutation + association

Rewrite C ∨ (A ∨ B) into (A ∨ B) ∨ C.

```
C ∨ (A ∨ B)
= ?
```

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Simplify Boolean expressions
State the simplified form of each expression:
(i) ¬¬A (ii) ¬A ∧ ¬B (iii) ¬(¬A ∧ ¬B)

Question 2

Simplify the popcorn expression
A cinema gives free popcorn when a voucher is shown and either a soft drink or a bottle of water is purchased.
E = voucher shown; F = soft drink bought; G = water bought.
Simplify (E ∧ F) ∨ (E ∧ G) .

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Cinema discount: Boolean expression
A discounted cinema ticket is issued if the customer is under 18 AND has a student card , OR is over 60 AND has proof of age .
A = under 18; B = student card; C = over 60; D = ID. Write the Boolean expression for the output Q (discount ticket).

Question 4

From a gate circuit to an expression
Two OR gates take inputs A, B and C, D respectively. Their outputs feed a third OR gate. Write the complete Boolean expression for the circuit output.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

De Morgan's Laws swap AND and OR while negating each term.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.